THE DEVELOPMENT OF THE INNOVATION OF SMART COIN-OPERATED DOCUMENT PRINTING SYSTEM USING RASPBERRY PI

A

Capstone Project

Presented to

The Faculty of the College of Computer Studies

University of Rizal System

Binangonan, Rizal

In Partial Fulfillment

of the Requirements for the Degree

Bachelor of Science in Information Technology

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November 2025

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ACKNOWLEDGMENT

First and foremost, to **GOD ALMIGHTY**, for providing strength, wisdom, and the resources necessary throughout the journey of this research.

The completion of this project would not have been possible without the support, guidance, and encouragement of many individuals. The proponents would like to take this opportunity to extend their heartfelt gratitude to all those who played a vital role in the successful realization of this project:

DR. JOANNE S. CONCEPCION, Campus Director of University of Rizal System Binangonan Campus, for constant encouragement and for the kind of academic environment which empowered us to grow. Her leadership has been the beacon of support throughout our studies;

PROF. ENRIQUE L. VILLONES, Campus Director University of Rizal System Cainta Campus and Capstone Project Adviser, whose vision and dedication inspired us to achieve excellence. His guidance had been a very essential part of our academic journey;

DR. JOY SG. CRUZ, Dean of College of Computer Studies of University of Rizal System Binangonan Campus for her encouragement and belief in our potential;

PROF. RUBY B. HERMANO, our Panel Chairman and Capstone Project Instructor, for her unwavering guidance, insightful suggestions, and constructive feedback that significantly contributed to the development and success of this study. Her commitment, support, and expertise have been invaluable throughout the entire process;

PROF. MELVIN A. SURIAGA, our Capstone Project IT Expert, for selflessly sharing his technical expertise which has really added to the actualization and fine tuning of our work. His knowledge has really added depth to our project;

PROF. MONETTE A. PILAPIL, our Critic Reader, for her thoughtful and critical feedback. Her careful perusal and suggestions have been very helpful in polishing the manuscript of this work;

Last but not the least, to our parents: **MR. AND MRS. BABANTE, MR. AND MRS. INTANO, MR. AND MRS. LAURENTE, MR. AND MRS. LUMBIS, MR. AND MRS. PANLUBASAN**, their sacrifices, love, and support have been the foundation on which our success will stand tall. The faith of our parents in us has encouraged us to continue being successful.

- **The Project Team**

DEDICATION

To our beloved parents — Mr. and Mrs. Babante, Mr. and Mrs. Intano, Mr. and Mrs. Laurente, Mr. and Mrs. Lumbis, and Mr. and Mrs. Panlubasan — your sacrifices, unconditional love, and unwavering support have been the foundation upon which our success stands tall. Your faith in us has motivated us to strive for excellence and persevere through every obstacle.

We also dedicate this research to the University of Rizal System – Cainta Campus, our second home, which has provided us not only with knowledge but also with the values and discipline that shaped us into responsible individuals. To the Office of the Student Development Services (OSDS), thank you for serving as the venue for this study and for your continued commitment to fostering student growth and innovation.

To God, to our families, and to our university — this accomplishment is as much yours as it is ours.

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TABLE OF CONTENTS

LIST OF FIGURES

Chapter 1

THE PROBLEM AND ITS BACKGROUND

Introduction

In the coming years, Information Technology (IT) will continue to serve as a driving force behind global progress, shaping the way industries operate and societies function. It will foster automation, enhance communication, and stimulate innovation across sectors such as business, education, healthcare, and governance. The rapid advancement of digital technologies—including artificial intelligence, embedded systems, and the Internet of Things (IoT)—will redefine efficiency, accessibility, and connectivity in both public and private institutions. As organizations increasingly depend on IT solutions for data processing, decision-making, and service delivery, technology will become indispensable in ensuring operational excellence and sustainability.

Within this global shift toward digitalization, the education sector will remain one of the primary beneficiaries of technological innovation. The integration of IT in schools and universities will transform traditional processes into automated and data-driven systems, improving academic efficiency and student engagement. These advancements will pave the way for smarter campuses that prioritize accessibility, convenience, and productivity through the use of affordable and sustainable technological solutions.

The Smart Coin-Operated Document Printing System will incorporate advanced technologies such as the Raspberry Pi 4B microcomputer, a coin acceptor module, wireless connectivity, a printer, and a locally hosted web server. The Raspberry Pi 4B will serve as the central processor that will manage communication between the hardware components, while the local host will handle file transfers and printing operations through a Wi-Fi network. These integrated technologies will enable users to upload documents from their mobile devices prior to coin verification and will initiate printing once payment has been confirmed. By integrating automated hardware with a web-based interface, the system will deliver a dependable, efficient, and fully autonomous printing service that will function seamlessly within the school premises.

The innovation of the Smart Coin-Operated Document Printing System will be developed to address the growing need for accessible, automated, and cost-efficient printing services within educational institutions. The Raspberry Pi will be selected due to its affordability, versatility, and compatibility with various components such as printers, sensors, and network modules. A local host will be implemented to allow offline functionality, ensuring that the system will remain operational even without internet connectivity. This technological innovation will demonstrate how automation and embedded systems will streamline daily activities while enhancing service efficiency. Through this project, students and faculty members will gain access to a reliable and convenient printing solution, thereby reducing time spent on off-campus printing tasks. The technology will benefit the institution by providing income-generating opportunities, promoting operational sustainability, and improving access to essential campus services.

The University of Rizal System – Cainta Campus, through its Office of the Student Development Services (OSDS), will oversee programs designed to promote student development, welfare, and leadership growth. It will supervise several departments, including Admissions, Scholarship, Guidance and Counseling, ID Production, Student Organizations, and OJT & Placement. Despite its vital role, OSDS will continue to encounter challenges in establishing sufficient income-generating services and accessible facilities that cater to student needs. One of the major issues will be the absence of on-campus printing facilities, which will compel students to seek services outside the university to print academic and organizational documents. This condition will cause inconvenience, additional expenses, and time consumption for students while depriving the university of potential revenue that could be utilized to support student programs and campus initiatives.

To address these challenges, the Smart Coin-Operated Document Printing System will be introduced as an automated, self-service solution that will enable students to print their documents conveniently within the university premises. Through the integration of the Raspberry Pi-based system with a coin acceptor and local hosting, the project will provide an efficient and user-friendly printing service that will operate independently without the need for internet connectivity or personnel assistance. This innovation will enhance the overall student experience by promoting convenience and affordability while supporting OSDS in generating additional income to fund student development programs. The project title, *"Innovation of Smart Coin-Operated Document Printing System Using Raspberry*

Pi,” will represent its primary objective—to integrate technology and practical application in delivering an automated, efficient, and revenue-generating service that will address both institutional and student requirements.

Objectives of the Project Study

To design and develop the Smart Coin-Operated Document Printing System Using Raspberry Pi that will provide users with an accessible, efficient, and automated method of printing documents without requiring continuous staff supervision. Additionally, the project will aim to establish an income-generating program that will assist the Office of Student Development and Services (OSDS) in supporting the school’s operational needs and funding essential student development activities.

Specifically, the study will aim to:

1. Evaluate the system’s performance through a quality assurance test plan, targeting at least a 95% passing rate in functionality and reliability.
2. Measure user acceptance of the system in terms of:
    2.1. Design;
    2.2. Ease of Use;
    2.3. Functionality
    2.4. Efficiency; and
    2.5. Reliability.

3. Provide the Office of Student Development and Services (OSDS) with a cost-effective and self-service printing system that will improve accessibility and support campus revenue generation.

Scope and Limitations of the Project Study

The study will focus on the development of the Smart Coin-Operated Document Printing System using Raspberry Pi for the Office of Student Development and Services (OSDS), located in Cainta, Rizal, during the academic year 2025–2026.

The project will involve the design and development of an automated, self-service printing device intended for students and instructors of the University of Rizal System – Cainta Campus. The system will allow users to print academic or organizational documents directly from their mobile devices through a local network interface. It will be implemented within the campus premises to provide an affordable and convenient printing service while generating additional income for the institution.

The system will be programmed to accept coins as a mode of payment and will proceed to print documents upon successful payment verification. A local web host will be utilized to handle file uploads and printing operations, ensuring functionality even without an internet connection. The study will cover system design, development, testing, and user evaluation to determine its functionality, reliability, and level of user acceptance.

However, the system will have certain limitations. It will only support one user at a time, meaning simultaneous printing requests will not be accommodated. The file size will be limited, restricting the upload of large documents. Additionally, the system will only allow the printing of a specific number of pages to maintain operational efficiency and prevent excessive use of resources such as paper and ink, which also helps prevent users from inserting too many coins unnecessarily. Despite these limitations, the system will still achieve its goal of providing a dependable, self-service, and income-generating printing facility for the campus community.

Significance of the Project Study

This study and the developed Smart Coin-Operated Document Printing System using Raspberry Pi for the Office of Student Development and Services (OSDS) will be beneficial to the following:

University of Rizal System – Cainta Campus (URS): The project shall give the campus an innovative and affordable printing solution that will enrich its service, enhance technological progress, and generate a secondary source of income for institutional growth.

Office of Student Development and Services (OSDS): The system shall be an extra income-generating service that will subsidize student programs and organizational activities and enhance operational effectiveness through automation.

Faculty: The initiative will give faculty members a more efficient and hassle-free means to print academic papers, lesson plans, and student outputs, enhancing productivity and workflow.

Students: The system will provide students with a more convenient, cheaper, and easier means to print documents across the campus, saving time and costs incurred through off-campus print services.

Future Researchers: The research will provide a worthwhile reference for future studies in automation, embedded systems, and campus technologies based on IoT, promoting innovation and real-world applications of low-cost computing.

Definition of Terms

The following terms are operationally and/or conceptually defined for better understanding at the study.

Automation - The ability of the system to print automatically after the user uploads a document and provides the necessary payment.

Coin Acceptor Module - Detects and verifies coins before initiating the printing process.

Database - A structured collection of data used to store user information, transaction history, and print records.

Document Printing System - The main configuration that prints uploaded documents automatically upon payment confirmation.

Embedded System: Integrates all electronic parts such as the Raspberry Pi, coin acceptor, and printer to function as a single unit.

GSM Module - Sends real-time notifications (e.g., SMS alerts) to users or administrators after a printing transaction.

IoT (Internet of Things) - Refers to the interconnection of devices like Raspberry Pi, printer, and sensors for data exchange and control.

Local Web Host - Provides a platform for users to upload documents and issue print commands without requiring an internet connection.

Microcontroller - The hardware component responsible for executing specific control operations in the system.

Prototype - The initial working model of the Smart Coin-Operated Printing Kiosk used for testing and evaluation.

Raspberry Pi 4B - Serves as the system's central processor, controlling the printer, coin acceptor, and other connected modules.

Smart System - Enables the kiosk to function autonomously with automated payment processing and document printing.

Thermal Printer - A type of printer that uses heat-sensitive paper to produce text or images efficiently.

User Interface - The screen or web page where users upload files, input payment, and initiate the printing process.

Web Server - Manages the local hosting environment, processing print requests and managing data transactions.

Wi-Fi Connectivity - Allows users to connect their devices to the system's network for document transfer and wireless printing.

Chapter 2

REVIEW OF RELATED LITERATURE AND PROJECT STUDY

Related Literature

Printing in Raspberry Pi 4

The "Printing in Raspberry Pi 4" project (2022) is about using the Raspberry Pi 4 to facilitate wireless connection printing, highlighting the benefits of mobility, ease of access, multifunctionality, convenience, and affordability. Wireless printers let one work effectively without being stuck in a fixed location, useful in businesses like retail, logistics, and healthcare where the need to print documents, receipts, or labels in an instant is required. For companies that employ mobile workers such as field inspectors or police officers, wireless communication facilitates on-the-spot printing of invoices, reports, and test results without going back to the office.

Wireless printers make availability better through the ability to print without Wi-Fi, with iOS and Android capabilities. With this capability, employees can print exam data, tickets, and labels efficiently with little disruption. They also have infrared technology to connect several devices and have the capacity to create good quality prints such as banners, receipts, and barcodes, providing functionality in commercial and professional environments.

The project points out that wireless printers are space-saving and portable, eliminating space requirements and enabling many users to be connected to one printer, hence lowering the cost of equipment. They are made to reduce wastage, particularly convenient in conferences or seminars where on-demand printing avoids unnecessary materials. Wi-Fi printers are also economical and suitable for small enterprises or new ventures because they are portable, have a long-lasting battery, and consume little power. The convergence of wireless printing with the Raspberry Pi 4 offers a useful and adaptive printing solution supported by Microsoft, macOS, and other Linux environments, showing potential for large-scale uses in personal and business settings.

Self-Service Print Kiosk

A self-service print kiosk, as described by Dyer (2024), is an advanced automated printing system that produces high-quality, large-format printed materials such as posters, signs, and maps directly to consumers. The kiosk would house a wide-format printer designed to handle large paper rolls, generally up to 24 × 36 inches, and mechanisms for cutting, cooling, rolling, and shrink-wrapping of the printed output in a secure enclosure. Its interior includes consumable storage for ink or toner, paper rolls, and packaging material to enable continuous operation and high-capacity printing with infrequent manual intervention. It uses an interactive user interface that allows users to browse through a centralized or remote image database, select options for printing, and complete the transaction through integrated payment processing. Finally, after printing, the kiosk dries or cures the ink or toner, rolls, and packages the output to prevent smudging and maintain professional-grade quality.

Operational efficiency is further improved by remote monitoring and automated diagnostics, which detect consumable lows, mechanical faults, and deviations in performance for rapid servicing to minimize kiosk downtime. The kiosk is designed as a fully enclosed, modular machine in which major functional units—the printer assembly, cooling mechanism, cutting module, and packaging system—can be replaced quickly by technicians to improve reliability and reduce the complexity of maintenance. Furthermore, the system connects to a cloud-based management platform to synchronize content libraries, track usage metrics, and update machine settings across multiple kiosk locations. High-speed production, real-time printing availability, and seamless end-to-end automation are emphasized in the design, making it suitable for practical application in on-demand printing situations within retail spaces, malls, theaters, and public areas with high traffic. Compared to traditional over-the-counter printing services, the print kiosk combines digital selection and automated printing processes with immediate product dispensing, hence reflecting considerable improvements in convenience, accessibility, and operational sustainability.

Swiss Post x KUARIO: Revolutionising Customer Experience with Modern Printing Solutions

In the study of Le (2024), the modernization of Swiss Post's document services demonstrates the transformative impact of automated, multi-functional printing kiosks on both operational efficiency and customer experience. Prior to the implementation of KUARIO's solution, Swiss Post used older, traditional coin-operated photocopy machines that were limited in their functionality, offering only black-and-white outputs and requiring labor-intensive, inefficient manual cash handling by employees. The decline in usage of these older machines encouraged the postal service to seek out more flexible technology-driven alternatives.

KUARIO's MFP kiosks integrated high-capacity printing hardware with large touch-screen interfaces. As a result of using this facility, customers print, scan, and copy materials in both color and black and white, using various paper sizes such as A3 and A4. The kiosks allowed users to upload documents directly from mobile devices or cloud services such as Google Drive, Office 365, and Dropbox, increasing access and convenience. Payment systems were diversified to accept debit and credit cards, local digital payment systems such as Twint, and coins to avoid human cash handling and minimize transactional errors.

The kiosks also featured cloud management dashboards with real-time monitoring of paper levels, consumables, and usage statistics, enabling employees to oversee operations efficiently without getting bogged down in routine maintenance. The deployment of KUARIO MFP kiosks across 320 Swiss Post branches saw usage increase by 89%, reflecting customer demand for flexible, user-friendly self-service printing. It's also emphasized in the case study that these kiosks help ensure operational sustainability by reducing labor costs, cutting errors, and smoothing workflows. This system, through the use of digital payments and cloud connectivity, future-proofs the services of Swiss Post while improving customer satisfaction, increasing the flow of customers in branches, and providing solutions that are scalable to adapt to evolving service demands in public and commercial environments.

Turning My Old Printer into a Wireless Printer with a Raspberry Pi

In the study of Tripad (2020), the capstone project “Turning My Old Printer into a Wireless Printer with a Raspberry Pi” focused on reengineering a vintage HP LaserJet 1020 printer into a fully operational wireless printing system using a Raspberry Pi 3B. The research sought to deliver an energy-efficient, cost-effective, and user-friendly method that allows direct printing from mobile and desktop devices without using a computer powered all the time. Through the use of Common Unix Printing System (CUPS) and Internet Printing Protocol (IPP), network-based sharing of printers was implemented by the researchers, which was reachable to Android and Windows clients alike, rendering Windows-based SAMBA configuration complex no longer necessary.

The project overcame major technical hurdles, including the installation of the Foo2zjs driver package in order to remain compatible with the HP LaserJet 1020 and resolving the USB connectivity and power supply problems, all in order to provide consistent and error-free printer functionality. In addition, the research focused on the optimization of the work area by minimizing the physical size of the printer and cable management, while continuing to provide maximum functionality for multiple users. As a whole, the project proved to be both practical and innovative in its method of modernizing aged printing machines, evidencing the feasibility of incorporating Raspberry Pi systems within normal printing environments to increase accessibility, convenience, and efficiency both at home and in the office.

HP digital printing press: Compute Module 4 makes monitoring and controlling industrial manufacturing processes easier and more efficient

The capstone project "HP Digital Printing Press" (2020) focused on revolutionizing industrial label printing by combining HP's next-generation LEPx (liquid electrophotographic) technology with a Raspberry Pi Compute Module 4–based control system to achieve increased speed, accuracy, and operational efficiency of the Indigo V12 Digital Press. The main challenge was to create a high-performance, small, and robust computing system that can thrive under challenging industrial conditions with dust, vibration, and around-the-clock uptime, as well as offer accurate control and responsive user interaction. HP installed six industrial-grade PPC-CM4-070 panel PCs per Indigo V12 unit, one for every impression station, with 1024×600 capacitive touchscreens for point-of-use monitoring and control, rugged I/O for sensor and industrial signal integration, and fanless rugged enclosures to resist environment stress. These panels were programmed with custom software using the Raspberry Pi ecosystem, enabling operators to perform real-time diagnostics, monitor printing operations, adjust settings, and synchronize workflows across multiple impression stations.

The project leveraged Raspberry Pi’s advantages in affordability, modular architecture, developer-friendly ecosystem, and long-term support to streamline system integration and reduce overall cost per unit while maintaining industrial-grade reliability. In comparison to other embedded solutions, Raspberry Pi provided software flexibility, a wide range of GPIOs, and scalability that enabled HP to put multiple units into a press without overrunning budget. Use of the platform greatly minimized wiring complexity, eased the operator interface, and compressed development timelines. The resulting Indigo V12 Digital Press printed at speeds of 120 linear meters per minute, a significant boost in digital label manufacturing capacity. Moreover, the system provided high-quality output, smooth workflow integration, and real-time system monitoring, demonstrating the applicability of scalable embedded computing solutions to future-generation industrial equipment. This research highlights the powerful integration of IoT-capable embedded systems with industrial production, to show how small, modular computer platforms such as Raspberry Pi Compute Module 4 can maximize efficiency, ease of use, and performance in high-volume print activity.

Related Project Studies

Optimizing User Interaction and Notification Efficiency in Smart Coin-Operated Printing Kiosk with Real-Time SMS Notifications and Interactive Features

In the study of Falcon, Saligue, R., and Saligue, E. (2024), the design and implementation of a smart coin-operated printing kiosk demonstrates how integrating real-time SMS notifications and interactive features can optimize user interaction and operational efficiency. This research was focused on creating a prototype kiosk that would grant students the convenience of printing documents anywhere on campus, with immediate feedback on system status and transaction completion through notifications enabled by GSM. The kiosk used a central processing unit combined with an Arduino microcontroller for managing user selection, coin validation, and print execution. For example, users could insert coins, select documents to be printed through an interactive touch-screen interface, and receive instantaneous SMS alerts in case of job completion or error notifications related to paper jams. The kiosk was integrated with cloud storage and databases that allow secure and scalable management of requests for printing, records of transactions, and logs from the system.

The study had several key features that were focused on the kiosk's functionality, such as the accuracy of a print job, overall effectiveness in satisfying users' needs, and learnability for first-time users. The interface allowed students to give their feedback or ask for assistance from the machine itself, and the SMS notification system reduces the latency of delay and enhances the reliability of service delivery. By combining a robust software framework with hardware components like coin slots, microcontrollers, and servers connected for data processing, the kiosk provided an all-in-one campus printing solution. Data is acquired and processed using modified Prototype Demo Forms and ISO 9241 usability metrics, and results of this study showed that the kiosk enhanced user satisfaction and reduced the need for students to leave campus for printing, with a scalable model for future deployments of smart, automated printing systems in educational facilities. The findings prove that smart, coin-operated kiosks can provide efficient, user-friendly, high-tech solutions for on-demand printing in localized environments.

Microcontroller Based Coin Operated Printer

In the study of Wego (2023), targeted the creation of a coin-operated printer where users can photocopy and print documents with coins, bills, or e-wallet payments. The system was designed for utilization in schools, offices, and public spaces in order to offer quick and convenient printing services. The system runs on the microcontroller linked to the coin acceptor and printer and is programmed to initiate the printing after the proper payment is inserted. The project is seeking to solve typical issues like the lack of local printing stores, excessive costs of printing, and delays, particularly during deadline seasons.

The system's technology employs microcontrollers such as Arduino or Raspberry Pi, which govern the coin acceptor and regulate printer operations. The kiosks themselves usually come equipped with simple screens, USB ports, and basic security options. The study's main aims are to make printing easy and instant, bring revenue to the machine owner through pay-per-print transactions, and make the system easy and easy to use, even for first-time clients.

The study scope includes the design of self-service printing kiosks meant for use in schools, libraries, and cafes to foster convenience, ease of use, and efficiency. Nonetheless, the study also identifies some of its limitations, including currency compatibility problems, space and design limitations, low user assistance, maintenance needs, and the need for initial high setup costs to effectively run the system.

Development of an Internet of Things Based Printer Vending Machine

In the study of Perucho and Diamante (2024), the development of an IoT-based printer vending machine depicts innovation in automating document printing services with improved convenience for users and reduced manual workload for operators. This research was intended to design, develop, and implement a coin-operated printing system capable of functioning autonomously using Arduino technology by integrating software and hardware to enable users to print documents on demand. It was designed to manage all operations, right from verifying the coin to processing the print job and notifying errors, thus minimizing the need to monitor continuously by a human operator. They followed the structured Waterfall SDLC: planning, analysis and design, prototyping, coding, testing, deployment, and maintenance. The backend ensured system reliability, performance efficiency, and security via cloud connectivity, database management, and real-time notifications in case of errors.

The usability evaluation of the printer vending machine was conducted by IT experts, printing business owners, and casual users according to ISO 25010 standards on functional suitability, performance efficiency, usability, reliability, security, and maintainability. It was found that the system works highly effectively in all assessed categories as “Very Useful”. This research also pointed out some practical improvements that can be undertaken, including introducing wireless printing, increasing the number of supported document formats, providing better user account management, and optimizing all processes concerning coin, paper, and ink refills. This research shows how IoT and embedded systems have the potential to create an affordable, autonomous, and scalable solution for printing to increase accessibility, reduce complexity in operations, and offer quality printing services in diverse user environments.

Based Pay-Per- PesoPrint System: An Internet of Things (IOT) for Document Printing Services Print Solution

In the study of Cadiz (2024), the PesoPrint System was developed as an Internet of Things (IoT) based pay-per-print solution designed to provide efficient, user-centric, and automated document printing service in Western Mindanao State University – ESU Pagadian Campus. The system employs IoT technology to enable convenient connectivity between the physical print station and digital interfaces for users to engage with a touch-screen interface to choose files from their flash drives, preview pages, estimate costs depending on paper size and quantity, and start print jobs. The payment process is automated via a bill and coin acceptor with change provision, while communication between the main software and payment hardware is provided via Arduino UNO so that it can handle accurate and efficient transactions.

A companion mobile app lets users remotely upload and book files, create reservation codes, and get notified about their print job status, with administrators able to view sales reports and system notifications for errors like paper jams or low ink levels. The PesoPrint System improves accessibility, effectiveness, and dependability of printing services to students, faculty, and staff, and reduces risks of legacy printing approaches, including document loss, exposure to malware through unsafe flash drives, and spotty handling of payments. The scope of the system is, at present, within the campus only, with mobile app capability limited to the Android operating system and functionality dependent on network connection and hardware stability.

Development and Acceptability of a Coin-Operated Smart Printing Machine

In the study of Aguanta, Alumisim, Perez, Consulta, Macarista, and Quinzon (2024), the capstone project Development and Acceptability of a Coin-Operated Smart Printing Machine was conducted at the University of Rizal System-Antipolo Campus to address challenges in accessing printing services in schools and public libraries. The researchers developed a Coin-Operated Smart Printing Machine that integrates Arduino Uno R3 as its microcontroller, a coin slot for payment, a coin hopper for dispensing change, a printer, and a mini desktop for the user interface. The smart printing system will automate the printing of documents in exchange for coins to be input into it, ensuring convenience, accessibility, and efficiency for students and faculty.

This study determines the acceptability of the developed prototype in terms of design, functionality, accuracy, security, and maintenance. The T-test statistical method was used to determine significant differences in perception between experts and end-users. Results showed that the machine was highly accepted with a grand mean of 4.81 from experts and 4.78 from end-users, indicating high satisfaction and usability. In addition, results showed that there was no significant difference in the perception between the experts and end-users, ensuring that the system was reliable and practical.

This study uses Henry Chesbrough's Open Innovation Theory as its conceptual framework, highlighting collaboration and the integration of external ideas in the development of technology-based solutions. Similarly, the feedback and iterative improvement supported the development to make the machine relevant and responsive to users' needs.

The study adopted an IPO framework guided by Coomb’s System Approach. In the input stage, hardware and software requirements were identified such as Arduino IDE, Microsoft Visual Studio, mini desktop, printer, and servo motor. The process stage covers designing, assembling, programming, and testing of the machine. The output stage was able to produce a fully functional Coin-Operated Smart Printing Machine that was evaluated for efficiency and user satisfaction.

The project’s scope was printing Word, PDF, and PowerPoint files via USB and Wi-Fi connectivity, using Arduino, Python, HTML, CSS, and Visual Basic. However, the paper size was limited to one, only single-sided printing was supported, and it did not include paper bill acceptance or any advanced print options such as duplex or color adjustment. Yet, despite this, the project succeeded in showing how locally developed and affordable smart systems could be used in order to increase access to printing services within the confines of academic institutions.

Chapter 3

METHODOLOGY

Project Development Framework

To facilitate a systematic and orderly process towards creating the Smart Coin-Operated Document Printing System, this research will take the adherence to a structured model that explicitly establishes the flow from resources and information to the final outputs. The model is intended to direct the researchers in determining the required inputs, planning and conducting the development process, and assessing the produced system. It will also guarantee that the system has the purpose of fulfilling the goals of offering a convenient, automated, and revenue-generating printing service for the University of Rizal System – Cainta Campus.

The Input-Process-Output (IPO) model will be used in this research to demonstrate the relationship between the resources needed, the development processes, and the desired outcomes of the project. The IPO model will act as a conceptual and visual roadmap for both the system development and research stages, demonstrating how inputs are transformed through processes to yield outputs in response to the needs of OSDS, faculty, and students.

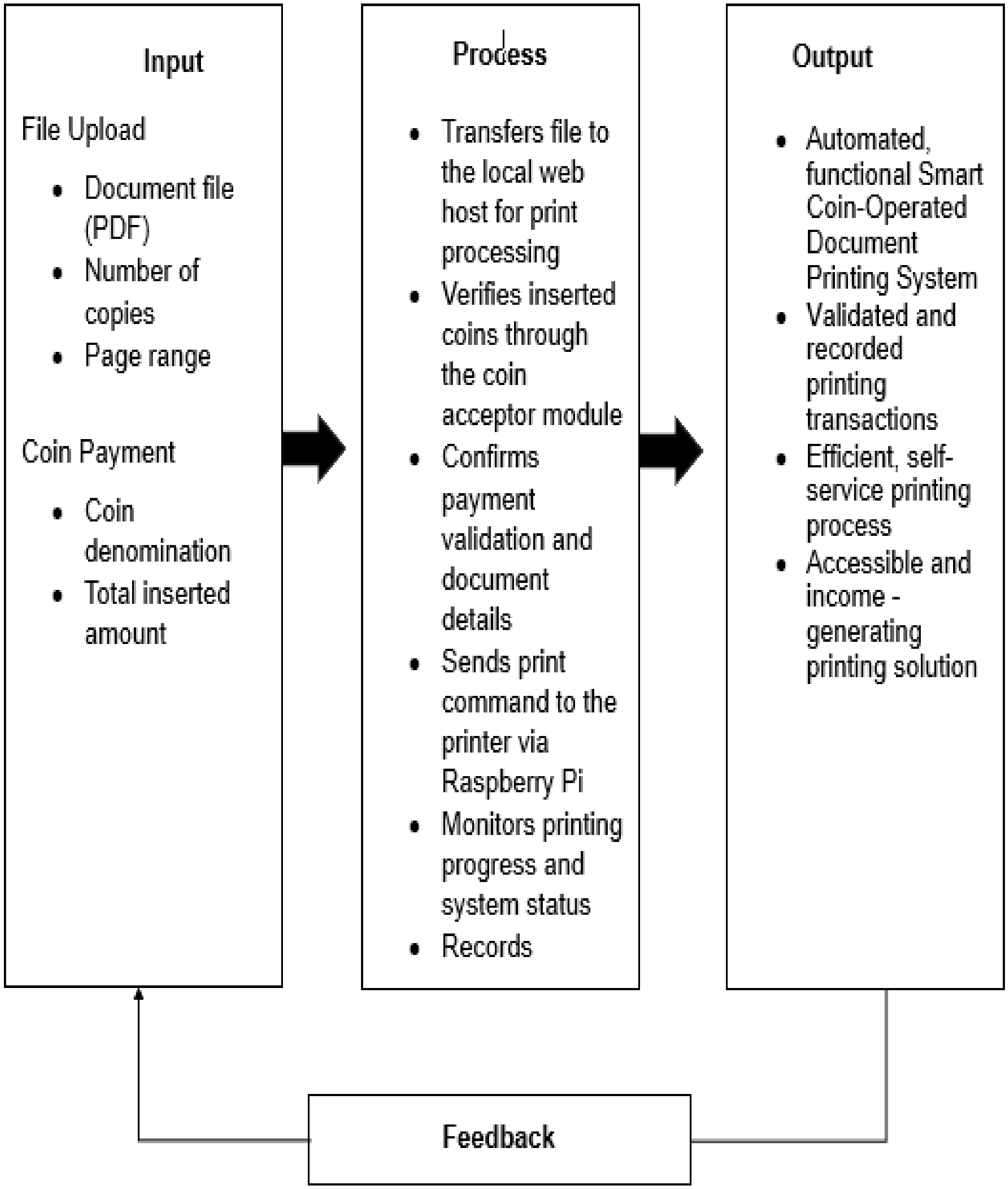

Figure 1

IPO Framework of the Innovation of Smart Coin-Operated Document Printing System Using Raspberry Pi

Figure 1 shows the IPO framework of the Innovation of Smart Coin-operated Document Printing System Using Raspberry Pi. The framework represents the sequential flow of data and activities that transform the inputs into meaningful outputs to achieve the project's objectives.

The Input comprises two main activities, namely, File Upload and Coin Payment. During file upload, the user will upload a document in PDF format, select the number of copies, and specify the page range to be printed. In the case of coin payment, coins of a particular denomination are fed in and the sum total of the fed-in amount is determined. These are the two key inputs required for executing the printing job.

Under the Process phase are a series of system-driven operations managed by the Raspberry Pi microcontroller. The file goes directly to the local web host for print processing once the document has been uploaded. The coin acceptor module verifies the validity of the inserted coins and ensures that the payment made corresponds to the cost required for printing. The system confirms payment validation and document details before sending the command to the connected printer for printing. During this process, the Raspberry Pi watches the progress of printing and the status of the system continuously; transactions are recorded for reference and accountability.

Finally, in the Output phase, an automated, functional Smart Coin-Operated Document Printing System is produced. The system ensures that all printing transactions are validated and recorded for efficient processing. This leads to a self-service, accessible, and income-generating printing solution that offers convenience for both students and staff in support of operational sustainability in the University of Rizal System-Cainta Campus.

The mechanism of Feedback acts as a continuous improvement component: it gathers both data from users and system performance logs, providing important building blocks for future developments to improve the overall experience and avoid potential issues.

Locale of the Project study

The research will be conducted at the Office of the Student Development Services (OSDS) of the University of Rizal System – Cainta Campus, which will be located at Gate 1, Karangalan Village, Brgy. San Isidro, Cainta, Rizal. Established in 2001, the OSDS will serve as the university's central office responsible for overseeing and implementing programs that will promote student development, engagement, and welfare. It will manage accredited student organizations, facilitate leadership and training activities, and ensure the holistic growth of students in both academic and non-academic aspects.

Since the OSDS will have direct interaction with the student body and will handle frequent document-related transactions such as registration, report submissions, and event proposals, it will serve as an ideal venue for the implementation and evaluation of the Smart Coin-Operated Document Printing System. The system will aim to enhance accessibility, efficiency, and convenience in document printing within the campus. Through this innovation, students and staff will experience a modernized self-service printing process that will support the university's vision of advancing technological development and promoting sustainable campus operations.

Figure 2

University of Rizal System – Cainta Campus Building

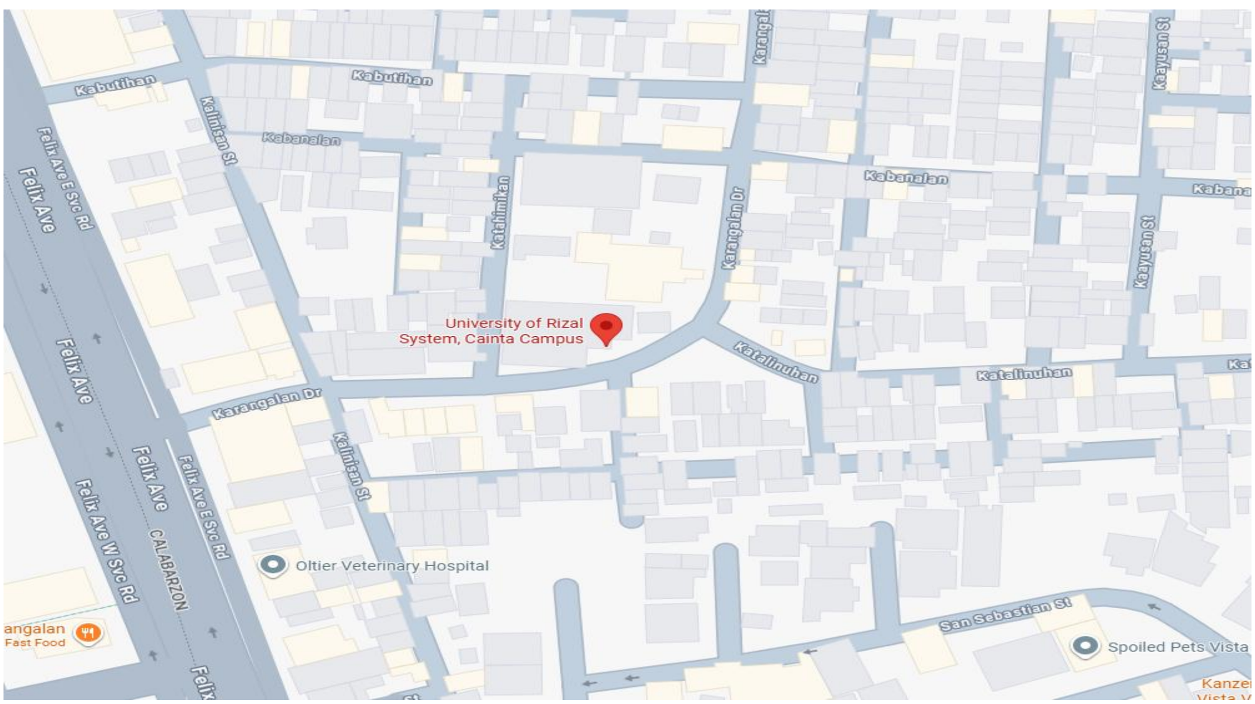

Figure 3

University of Rizal System – Cainta Campus Vicinity Map

Subject of the Project Study

This study will involve a total of twenty (20) respondents from the University of Rizal System – Cainta Campus, consisting of five (5) faculty or staff members, and fifteen (15) students. These respondents were selected because they represent the primary users of the Smart Coin-Operated Document Printing System. Their insights and feedback will be valuable in assessing the system’s usability, functionality, and overall effectiveness in providing convenient, automated, and accessible printing services within the campus.

Procedure of the Project Study

A capstone project that is successful will illustrate systematic planning and scientific implementation. To realize this, the researchers will employ a systematic and sequenced process in carrying out the study.

The process would start with meeting the capstone project instructor to discuss the scope, aim, and requirements of the study. The researchers will proceed with brainstorming sessions to come up with probable project titles and ideas afterwards. Interviews and casual conversations will be held with students, instructors, and OSDS personnel for a better understanding of certain issues and problems connected to document printing in the campus.

From the collected data, the researchers will formulate a number of proposed system titles and present them for evaluation by the panel through title defense. The chosen title—"The Innovation of Smart Coin-Operated Document Printing System Using Raspberry Pi"—will then be the basis of the study. After title approval, the group will jointly prepare the first few parts of the manuscript that consist of the Introduction, Objectives of the Study, Scope and Limitations, Significance of the Study, Review of Related Literature and Studies, and Proposed Methodology. After these chapters are ready, the manuscript will be presented to the project adviser and a critic reader, who will give comments and suggestions for improvement. The researchers will then revise the document accordingly, making it academically sound and complete. Lastly, the enhanced manuscript will be submitted to the colloquium panel for review and additional suggestions.

Once all suggestions have been incorporated, the researchers will edit, finalize, and make a soft-bound version of the capstone project that is to be submitted to the capstone project instructor for final approval and review.

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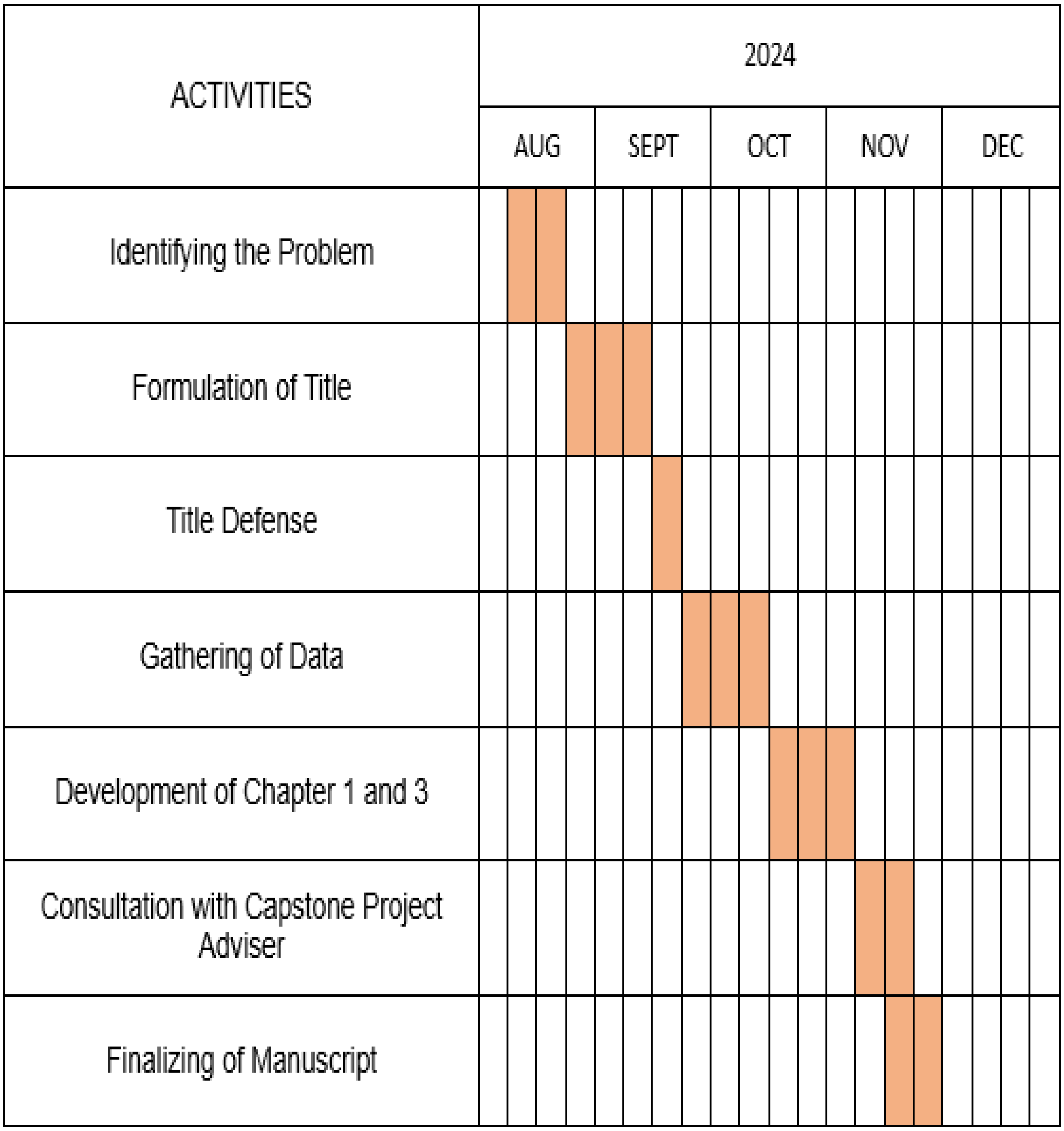
APPENDICES

Appendix A

Gantt Chart of Activities



Appendix B

Project Team Assignment Form

Republic of the Philippines
UNIVERSITY OF RIZAL SYSTEM
Province of Rizal
Cainta Campus

College of Industrial Technology
Bachelor of Science in Information Technology

URS Cainta Campus Tel: (02) 359-8776 / 532-0484 / 703-5871

Project Team Assignment Form

Group No: 6

Name	Role	Signature	Email Address	Mobile Number
James Kyle S. Laurente	Project Manager		jameskylelaurente555@gmail.com	09153622341
Jhon Rick C. Babante	Quality Analyst		babantejhonrick@gmail.com	09918681406
Jerald D. Intano	System Analyst		intanojerald@gmail.com	09913218606
John Gerome E. Lumbis	Programmer		lumbisgerome566@gmail.com	09922256852
Greg Mark B. Panlubasan	Technical Documenter		panlubasangreg@gmail.com	09674173191

Approved by:

Ruby B. Hermano
Subject Professor

Appendix C

Approved Project Form

Republic of the Philippines
UNIVERSITY OF RIZAL SYSTEM
Province of Rizal
Cainta Campus

College of Computer Studies

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

APPROVED PROJECT TITLE FORM

The Proponents:

NAME AND SIGNATURE	ROLE
James Kyle S. Laurente	Project leader
John Gerome E. Lumbis	Programmer
Jerald D. Infano	System Analyst
Greg Mark B. Panlubasan	Technical writer
Jhon Rick C. Babante	Quality Analyst

APPROVED PROJECT TITLE:

The innovation of smart coin-operated document printing system using raspberry pi

Recommending Approval:

Approved by:

RUBY B. HERMANO
Capstone Project Instructor

Noted by:

Joy SG. Cruz, PhD
Dean, College of Computer Studies

Appendix D

Letter for Conducting Study

Republic of the Philippines
UNIVERSITY OF RIZAL SYSTEM
Province of Rizal
Cainta Campus

College of Computer Studies

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

September 10, 2025

Mrs. Marjorie DF. San Juan
OSDS Coordinator
University of Rizal System Cainta
Gate 1 Karangalan Village, Brgy. San Isidro Cainta, Rizal

Dear Sir\Madam,

Greetings of Peace!

We are currently conducting a capstone project entitled, "The Development of the Innovation of Smart Coin-Operated Document Printing System Using Raspberry Pi" as partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology.

In connection with this, may we request from your good office to allow us to conduct the study in University of Rizal System Cainta, located at Gate 1 Karangalan Village, Brgy. San Isidro Cainta, Rizal, on September 10, 2025, at 10:00 AM. Rest assured that the data will be treated with outmost confidentiality.

We fervently hope for your kindest approval.

Thank you very much and God speed.
Very truly yours,

The Researchers

Member's name and signature over printed name

Jhon Rick C. Babante

Jerald D. Pitano

James Kyle S. Laurente

John Gerome E. Lumbis

Greg Mark B. Panlubasan

PROF. ENRIQUE L. VILLONES
Adviser (with signature over printed name)

Noted by:

RUBY B. HERMANO
Subject Instructor

Recommending Approval:

MARJORIE DF. SAN JUAN, MA.Ed	**RAFAEL GERSON G. ERIVE**
OSDS Coordinator	CIT – Program Head

Approved by:

ENRIQUE L. VILLONES, MAT
Campus Director

Republic of the Philippines
UNIVERSITY OF RIZAL SYSTEM
Province of Rizal
Cainta Campus

College of Industrial Technology

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

DATE: November 25, 2025

PROF. ENRIQUE L. VILLONES
Campus Director
This Campus

Dear Prof. Villones

In consideration of your qualifications in the field of research, upon recommendations of our Research Professor we would like to ask for your approval to be our **PROJECT ADVISER.**

As part of being our **ADVISER** stated in the **CAPSTONE PROJECT**, may we ask your help on the following matters:

1. Meets the team regularly (as per scheduled, NOTE: the team must seek proper appointment) to answer questions and help resolve issues and conflicts.
2. Points out errors in the development work, in the analysis, or in the documentation. The adviser must remind the Proponents to do their work properly.
3. Reviews thoroughly all deliverables at every stage of the Capstone Project, to ensure that they meet the college's standards.

The students who shall be under your scrutiny include

Name	Course	Signature
1. Jhon Rick C. Babante	BSIT	
2. Jerald D. Intano	BSIT	
3. James Kyle S. Laurente	BSIT	
4. John Gerome E. Lumbis	BSIT	
5. Greg Mark B. Panlubasan	BSIT	

This project proposal is entitled: **<u>THE DEVELOPMENT OF THE INNOVATION OF SMART COIN-OPERATED DOCUMENT PRINTING SYSTEM USING RASPBERRY PI</u>**

Thank you for your usual support to the research program/s of the College.

Very truly yours

Conformed:

PROF. ENRIQUE L. VILLONES
DATE: NOVEMBER 26, 2025

RUBY B. HERMANO
Subject Instructor

Republic of the Philippines
UNIVERSITY OF RIZAL SYSTEM
Province of Rizal
Cainta Campus

College of Industrial Technology

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

DATE: November 25, 2025

PROF. MELVIN A. SURIAGA
Faculty
This Campus

Dear Prof. Suriaga

In consideration of your qualifications in the field of research, upon recommendations of our Research Professor we would like to ask for your approval to be our **PROJECT IT EXPERT.**

As part of being our **IT EXPERT** stated in the **CAPSTONE Project**, may we ask your help on the following matters:

1) Validate the endorsement of the adviser. The panel serves as "Internal
2) Auditors", putting some form of check and control on the kinds of Capstone Projects being approved by the College.
3) Evaluate the deliverables.
4) Recommend a verdict.
5) Listen and consider the request of the adviser and/or the Proponents.
6) Nominate a Capstone Project for the Outstanding Capstone Project Award.

The students who shall be under your scrutiny include

Name	Course	Signature
1. Jhon Rick C. Babante	BSIT	
2. Jerald D. Intano	BSIT	
3. James Kyle S. Laurente	BSIT	
4. John Gerome E. Lumbis	BSIT	
5. Greg Mark B. Panlubasan	BSIT	

The project proposal is entitled: **THE DEVELOPMENT OF THE INNOVATION OF SMART COIN-OPERATED DOCUMENT PRINTING SYSTEM USING RASPBERRY PI**

Thank you for your usual support to the research program/s of the College.

Very truly yours

Conformed:

PROF. MELVIN A. SURIAGA
DATE: NOVEMBER 25, 2025

RUBY B. HERMANO
Subject Instructor

Republic of the Philippines
UNIVERSITY OF RIZAL SYSTEM
Province of Rizal
Cainta Campus

College of Computer Studies

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

DATE: November 25, 2025

PROF. MONETTE A. PILAPIL
Faculty/CBA Coordinator
This Campus

Dear Prof. Pilapil

In consideration of your qualifications in the field of research, upon recommendations of our Research Professor we would like to ask for your approval to be our **CRITIC READER.**

As part of being our Critic Reader, may we ask your help to review our thesis entitled: **THE INNOVATION OF SMART COIN-OPERATED DOCUMENT PRINTING SYSTEM USING RASPBERRY PI** against the set of structural rules that govern the composition of *sentences*, *phrases*, and *words* in the English language.

The students who shall be under your scrutiny include

Name	Course	Signature
1. Jhon Rick C. Babante	BSIT	
2. Jerald D. Intano	BSIT	
3. James Kyle S. Laurente	BSIT	
4. John Gerome E. Lumbis	BSIT	
5. Greg Mark B. Panlubasan	BSIT	

This project proposal is entitled: **THE DEVELOPMENT OF THE INNOVATION OF SMART COIN-OPERATED DOCUMENT PRINTING SYSTEM USING RASPBERRY PI**

Thank you for your usual support to the research program/s of the College.

Very truly yours

Conformed:

PROF. Monette A. Pilapil
DATE: November 24, 2025

RUBY B. HERMANO
Subject Instructor

CURRICULUM VITAE

JHON RICK C. BABANTE

Address: Block 1 Lot 43 Sitio Dalig 3, Barangay Dalig, Antipolo City

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EDUCATIONAL BACKGROUND

Attended	**Name of School**	**Year**
Tertiary	University of Rizal System – Cainta Campus	2023 - Present
	Course: BS Information Technology	
Secondary		
Senior High School	San Jose National High School	2021 - 2023
Junior High School	San Jose National High School	2015 - 2020
Elementary	Isaias S. Tapales Elementary School	2009 - 2015

SEMINARS OR TRAININGS ATTENDED

BITSIDE Web Dev (FTF training)	2023 - 2024
TechTalks (Online Seminar)	March 14, 2024
The Power of Data Analytics in IT Seminar (FTF)	April 2, 2025
Computer System Servicing NC II (TESDA**)**	July 31, 2025
Cyber-Ready	October 24, 2025

ORGANIZATION'S AFFILIATED

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Builders of Information Technology Society (BITS)
Inclusive date of Membership: 2023 – present

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Email: intanojerald@gmail.com

Contact Number: +639913218608

EDUCATIONAL BACKGROUND

Attended	**Name of School**	**Year**
Tertiary	University of Rizal System – Cainta Campus	2023 - Present
	Course: BS Information Technology	
Secondary		
Senior High School	Our Lady of Fatima University - Antipolo Campus	2021 - 2023
Junior High School	Antipolo National High School	2020 - 2021
	Don Antonio de Zuzuarregui Senior Memorial Academy	2017 - 2020
Elementary	Children's Garden School of Antipolo	2010 - 2017

SEMINARS OR TRAININGS ATTENDED

BITSIDE Web Dev (FTF training)	2023 - 2024
TechTalks (Online Seminar)	March 14, 2024
The Power of Data Analytics in IT Seminar (FTF)	April 2, 2025
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Cyber-Ready	October 24, 2025

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Inclusive date of Membership: 2023 – present

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EDUCATIONAL BACKGROUND

Attended	**Name of School**	**Year**
Tertiary	University of Rizal System – Cainta Campus	2023 - Present
	Course: BS Information Technology	
Secondary		
Senior High School	Sumulong College of Arts and Sciences - Antipolo Campus	2020 - 2022
Junior High School	San Roque National High School	2015 - 2020
Elementary	Nazarene Ville Elementary School	2010 - 2015

SEMINARS OR TRAININGS ATTENDED

BITSIDE Web Dev (FTF training)	2023 - 2024
TechTalks (Online Seminar)	March 14, 2024
The Power of Data Analytics in IT Seminar (FTF)	April 2, 2025
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EDUCATIONAL BACKGROUND

Attended	Name of School	Year
Tertiary	University of Rizal System – Cainta Campus	2023 - Present
	Course: BS Information Technology	
Secondary		
Senior High School	Antipolo City Senior High School	2020 - 2022
Junior High School	Antipolo National High School	2015 - 2020
Elementary	Sta. Cruz Elementary School	2010 - 2015

SEMINARS OR TRAININGS ATTENDED

BITSIDE Web Dev (FTF training)	2023 - 2024
TechTalks (Online Seminar)	March 14, 2024
JAVA Programming NC III (TESDA)	2024 - 2025
The Power of Data Analytics in IT Seminar (FTF)	April 2, 2025
Computer System Servicing NC II (TESDA**)**	July 31, 2025
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Contact Number: +639674173191

EDUCATIONAL BACKGROUND

Attended	**Name of School**	**Year**
Tertiary	University of Rizal System – Cainta Campus **Course:** BS Information Technology	2023 - Present
Secondary		
Senior High School	Our Lady of Fatima University - Antipolo Campus	2021 - 2023
Junior High School	Mambugan National High School	2017 - 2021
Elementary	Palid II elementary School	2010 - 2017

SEMINARS OR TRAININGS ATTENDED

BITSIDE Web Dev (FTF training)	2023 - 2024
TechTalks (Online Seminar)	March 14, 2024
The Power of Data Analytics in IT Seminar (FTF)	April 2, 2025
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Inclusive date of Membership: 2023 – present